ARITHMETIC SHADOWS OF CHIRAL ALGEBRAS

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Abstract. The ordered bar complex of a chiral algebra $\mathcal{A}$ on a family of elliptic curves produces genus-1 amplitudes whose Fourier coefficients are divisor sums. The Dirichlet series formed from these coefficients factors as $D_2(\mathcal{A}, s) = -24\kappa(\mathcal{A})\,\zeta(s)\,\zeta(s-1)$: a product of Riemann zeta functions, scaled by the modular characteristic $\kappa(\mathcal{A})$. We develop this observation into a systematic arithmetic theory of chiral algebras. For lattice vertex algebras, a constrained Epstein zeta function $\varepsilon_s^c(V_\Lambda)$ restricted to the bar sublattice recovers the Riemann zeta function at rank 1 and decomposes into Hecke L-functions at higher rank. The shadow L-function $L_{\mathcal{A}}^{\mathrm{sh}}(s) = \sum_{r\geq 2} S_r\, r^{-s}$, built from shadow tower coefficients, is stratified by depth class, with poles at $s = 1, 2$ and trivial zeros at negative integers. At integer level, ordered chiral homology recovers the Verlinde formula; the Verlinde dimensions are polynomials $P_g(n) = n^{g-1}(n^2-1)\,R_{g-2}(n^2)$ whose leading coefficients are values $\zeta(2g-2)/(2^{g-2}\pi^{2g-2})$ of the Riemann zeta function. At genus 2, the Maurer–Cartan element connects to Siegel modular forms via the Böcherer conjecture. Every construction begins with the Heisenberg algebra and escalates by Lie-theoretic complexity.

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1. Introduction

1.1. **The arithmetic content of the shadow tower.** The bar complex of a chiral algebra $\mathcal{A}$ on an algebraic curve X is a cochain complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct and differential encoding the OPE through the logarithmic kernel $\eta = d\log(z_1 - z_2)$. Over a family of genus-g curves, the fiberwise differential squares to $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, where $\omega_g = c_1(\lambda)$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_g$. The single scalar $\kappa(\mathcal{A})$, the *modular characteristic*, controls the obstruction to flatness of the bar complex as a family.

The shadow obstruction tower $\{S_r(\mathcal{A})\}_{r\geq 2}$ extracts the degree-r constant terms of the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$, with $S_2 = \kappa$. The shadow coefficients are rational functions of the central charge and level, carrying algebraic data from the OPE into the tower.

The question this paper addresses: do these rational numbers carry arithmetic meaning beyond their algebraic origin?

1.2. **The Heisenberg base point.** The Heisenberg algebra $\mathcal{H}_k$ is the abelian Kac–Moody algebra at level k, with OPE $J(z)J(w) \sim k/(z-w)^2$ and modular characteristic $\kappa(\mathcal{H}_k) = k$. The genus-1 amplitude of $\mathcal{H}_k$ over the family $\mathcal{C}_\tau \to \mathfrak{H}$ is

$$F_1^{\mathrm{conn}}(\mathcal{H}_k; q) = -k \sum_{n\geq 1} \sigma_1(n)\, q^n = -k \cdot \frac{E_2^*(\tau) - 1}{24}, \tag{1.1}$$

where $E_2^*(\tau) = 1 - 24\sum_{n\geq 1}\sigma_1(n)\, q^n$ is the quasi-modular Eisenstein series and $\sigma_1(n) = \sum_{d|n} d$ is the sum-of-divisors function. The Fourier coefficients are divisor sums: the simplest arithmetic functions in analytic number theory.

Every construction in this paper begins with the Heisenberg computation and ascends to the affine Kac–Moody, Virasoro, $\mathcal{W}_N$, and lattice families by increasing the Lie-theoretic complexity. The arithmetic becomes richer at each step, but the Heisenberg case already contains the essential structure.

1.3. **Summary of results.** The paper is organised as follows.

Section 2 proves the shadow Eisenstein theorem: the Dirichlet series $D_2(\mathcal{A}, s)$ formed from the genus-1 Fourier coefficients factors as a product of Riemann zeta functions scaled by $\kappa(\mathcal{A})$.

Section 3 constructs the categorical zeta function $\varepsilon_s^c(V_\Lambda)$ for lattice vertex algebras by restricting the Epstein series to the bar sublattice, and proves that it recovers the Riemann zeta function at rank 1.

Section 4 studies the shadow L-function $L^{\mathrm{sh}}_{\mathcal{A}}(s)$, proves its analytic properties (poles at $s = 1, 2$, trivial zeros at negative integers), and shows that the depth decomposition of the shadow tower induces a stratification of $L^{\mathrm{sh}}_{\mathcal{A}}$ by depth class.

Section 5 recovers the Verlinde formula from ordered chiral homology at integer level, proves the Verlinde polynomial family $P_g(n) = n^{g-1}(n^2 - 1)\, R_{g-2}(n^2)$ through genus 6, and identifies the leading coefficient as a zeta value.

Section 6 connects the genus-2 Maurer–Cartan element to Siegel modular forms via the Saito–Kurokawa lift and the Böcherer conjecture.

Section 7 develops the constrained Epstein zeta function, its Hecke decomposition, and the Dirichlet–sewing lift.

1.4. **Conventions.** All chiral algebras are $\mathbb{Z}$-graded with cohomological convention ($|d| = +1$). The modular characteristic is

$$\kappa(\mathcal{A}) := \begin{cases} k & \text{Heisenberg } \mathcal{H}_k, \\ \dfrac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee} & \text{affine Kac–Moody } V_k(\mathfrak{g}), \\ c/2 & \text{Virasoro } \mathrm{Vir}_c, \\ c\,(H_N - 1) & \text{principal } \mathcal{W}_N, \end{cases}$$

where $H_N = \sum_{j=1}^N 1/j$ is the N-th harmonic number. We write $q = e^{2\pi i\tau}$ for the nome. The ordered bar complex is $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and $|s^{-1}v| = |v| - 1$ is the desuspension.

2. The shadow Eisenstein theorem

2.1. **Genus-1 amplitudes from the bar complex.** Let $\mathcal{A}$ be a chirally Koszul algebra with modular characteristic $\kappa = \kappa(\mathcal{A})$. The bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ over the universal family of elliptic curves $\mathcal{C}_\tau \to \mathfrak{H}$ produces the genus-1 connected free energy

$$F_1^{\mathrm{conn}}(\mathcal{A}; q) = -\log\det(1 - K_q), \tag{2.1}$$

where K_q is the Bergman sewing kernel on the torus $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$, a trace-class operator whose Fredholm determinant replaces Polyakov's spectral-zeta regularisation of the Laplacian. The Heisenberg one-particle reduction expresses this as a q-series.

The genus-1 shadow amplitude is

$$\mathrm{Sh}_2^{(1)}(\mathcal{A},\tau) = \kappa(\mathcal{A}) \cdot E_2^*(\tau), \tag{2.2}$$

where $E_2^*(\tau) = 1 - 24\sum_{n\geq 1}\sigma_1(n)\,q^n$ is the quasi-modular Eisenstein series (weight 2, with the non-holomorphic completion $\hat{E}_2(\tau) = E_2^*(\tau) - 3/(\pi\,\mathrm{Im}(\tau))$ absorbing the anomaly from $d\log E(z,w)$). The Fourier coefficients $\sigma_1(n) = \sum_{d|n} d$ are the sum-of-divisors function.

Example 2.1 (Explicit genus-1 amplitudes)**.** For the standard families:

(a) *Heisenberg* $\mathcal{H}_k$*:* $\kappa(\mathcal{H}_k) = k$; $\mathrm{Sh}_2^{(1)} = k \cdot E_2^*(\tau)$.
(b) *Affine Kac–Moody* $V_k(\mathfrak{g})$*:* $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$; $\mathrm{Sh}_2^{(1)} = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)\cdot E_2^*(\tau)$. At $k=0$: $\kappa = \dim(\mathfrak{g})/2$; the amplitude does not vanish (the abelian limit is Heisenberg, not trivial). At $k=-h^\vee$: $\kappa = 0$; the amplitude vanishes (critical level).
(c) *Virasoro* Vir_c*:* $\kappa(\mathrm{Vir}_c) = c/2$; $\mathrm{Sh}_2^{(1)} = (c/2)\cdot E_2^*(\tau)$. Self-dual at $c=13$, where $\kappa+\kappa' = 13$.

2.2. **The Dirichlet series of divisor sums.** The Fourier coefficients $\sigma_1(n)$ assemble into a Dirichlet series with a classical factorisation.

Theorem 2.2 (Shadow Eisenstein theorem)**.** *Let $\mathcal{A}$ be a chirally Koszul algebra with modular characteristic $\kappa(\mathcal{A})$. The Dirichlet series*

$$D_2(\mathcal{A},s) := \sum_{n\geq 1}\sigma_1(n)\,n^{-s} \tag{2.3}$$

satisfies

$$D_2(\mathcal{A},s) = \zeta(s)\,\zeta(s-1), \tag{2.4}$$

where ζ is the Riemann zeta function. The genus-1 amplitude Dirichlet series is therefore

$$-24\,\kappa(\mathcal{A})\cdot\zeta(s)\cdot\zeta(s-1)\,: \tag{2.5}$$

an Eisenstein L-function with the modular characteristic as the sole chiral-algebra input.

Proof. The identity $\sigma_1(n) = \sum_{d|n} d$ is a Dirichlet convolution: $\sigma_1 = \mathrm{id} * \mathbf{1}$, where $\mathrm{id}(n) = n$ and $\mathbf{1}(n) = 1$. The Dirichlet series of a convolution is the product of the constituent Dirichlet series:

$$\sum_{n\geq 1}\sigma_1(n)\,n^{-s} = \Big(\sum_{n\geq 1} n^{1-s}\Big)\Big(\sum_{n\geq 1} n^{-s}\Big) = \zeta(s-1)\,\zeta(s).$$

The factor $-24\,\kappa(\mathcal{A})$ comes from the relation (2.2): the genus-1 amplitude has Fourier coefficient $-24\,\kappa(\mathcal{A})\,\sigma_1(n)$ at q^n. □

Remark 2.3 (Analytic properties). The product $\zeta(s)\,\zeta(s-1)$ has poles at $s=1$ (from $\zeta(s)$) and $s=2$ (from $\zeta(s-1)$), a functional equation inherited from those of the individual zeta factors, and zeros at the nontrivial zeros of ζ shifted by 0 and 1. The residue at $s=2$ is $\zeta(1^+)=\infty$; the regularised residue is $\mathrm{Res}_{s=2}\,\zeta(s)\zeta(s-1)=\zeta(2)=\pi^2/6$, giving a direct bridge from the modular characteristic to the value $\zeta(2)$.

Remark 2.4 (Non-automorphicity of the shadow tower). The shadow L-function $L^{\mathrm{sh}}_{\mathcal{A}}(s)=\sum_{r\geq 2} S_r\, r^{-s}$, built from the shadow tower coefficients S_r, is a different object from $D_2(\mathcal{A},s)$. It is *not* in the Selberg class: the coefficients S_3, S_4 violate the Ramanujan bound for class-M algebras, and $L^{\mathrm{sh}}_{\mathcal{A}}$ has no Euler product (the shadow coefficients are not multiplicative). The Riemann zeta enters through the genus-1 Eisenstein channel; the shadow tower itself carries arithmetic that is not automorphic.

2.3. **The Polyakov correspondence.** The shadow Eisenstein theorem has a physical counterpart. Polyakov's programme computes the string partition function by gauge-fixing the worldsheet metric to conformal gauge, producing a conformal anomaly $\langle T^a{}_a\rangle=(c/12)\,R$ and a ghost system with $c_{\mathrm{ghost}}=-26$. Every element of this programme has a bar-cobar counterpart.

Polyakov	**Bar-cobar**
Conformal anomaly $(c/12)\,R$	Modular characteristic $\kappa(\mathcal{A})$
$\det'_\zeta\,\Delta$ (Laplacian)	Fredholm $\det(1-K)$ (Bergman kernel)
Critical dimension $c=26$	$\kappa(\mathrm{Vir}_{26-c})=0$
Ghost system bc	Koszul dual $\mathcal{A}^!$
$F_g=\kappa\,\lambda_g^{\mathrm{FP}}$ (uniform)	Genus-g shadow amplitude

The genus-1 functional determinant $\det'_\zeta\,\Delta=|\eta(\tau)|^4\,(\mathrm{Im}\,\tau)^2$ is the Heisenberg contribution; the divisor sums in E_2^* encode the one-loop corrections. The shadow Eisenstein theorem extracts the multiplicative structure of these corrections.

3. The categorical zeta function

3.1. **Lattice vertex algebras and the bar sublattice.** Let Λ be an even lattice of rank r with inner product $\langle\cdot,\cdot\rangle$, and let V_Λ be the associated lattice vertex algebra. The vertex algebra V_Λ decomposes as $V_\Lambda=\mathcal{H}^{\otimes r}\otimes\mathbb{C}[\Lambda]$, where $\mathcal{H}^{\otimes r}$ is the rank-r Heisenberg algebra and $\mathbb{C}[\Lambda]$ is the group algebra. The modular characteristic is $\kappa(V_\Lambda)=r/2$.

The bar complex $\bar{B}^{\mathrm{ord}}(V_\Lambda)$ restricted to the lattice sector produces a sublattice $\Lambda^{\mathrm{bar}}\subset\Lambda$ consisting of nonzero lattice vectors that survive the augmentation: $\Lambda^{\mathrm{bar}}=\Lambda\setminus\{0\}$.

Definition 3.1 (Constrained Epstein zeta function)**.** For a lattice vertex algebra V_Λ of rank r, the *constrained Epstein zeta function* is

$$\varepsilon_s^c(V_\Lambda) := \sum\nolimits'_{\ell\in\Lambda^{\mathrm{bar}}} \|\ell\|^{-2s} = \sum_{\substack{\ell\in\Lambda\\ \ell\neq 0}} \langle \ell,\ell\rangle^{-s}, \tag{3.1}$$

where the prime denotes omission of $\ell = 0$.

This is the Epstein zeta function of the quadratic form $Q(\ell) = \langle \ell,\ell\rangle$ restricted to the bar sublattice. For a general even lattice, the Epstein zeta function $E_\Lambda(s) = \sum_{\ell\neq 0} Q(\ell)^{-s}$ is a classical object in analytic number theory, studied by Epstein [4, 5] and Siegel [9]. The constraint to Λ^{bar} is the bar-complex restriction: the augmentation ideal removes the vacuum, and the surviving lattice vectors carry the OPE data.

3.2. **Recovery of the Riemann zeta function.**

Theorem 3.2 (Rank-1 recovery)**.** *For the rank-1 lattice $\Lambda = \mathbb{Z}$ at self-dual radius $R = 1$ (i.e., the inner product $\langle m,n\rangle = 2mn$), the constrained Epstein zeta function is*

$$\varepsilon_s^c(V_\mathbb{Z}) = 2\cdot(2)^{-s}\sum_{m\geq 1} m^{-2s} = 2^{1-s}\,\zeta(2s). \tag{3.2}$$

At self-dual radius with the normalisation $\langle m,n\rangle = mn$: $\varepsilon_s^1(R{=}1) = 4\,\zeta(2s)$. The simplest lattice vertex algebra's constrained Epstein zeta is the Riemann zeta function (up to a power of 2).

Proof. At rank 1 with $\Lambda = \mathbb{Z}$ and quadratic form $Q(m) = m^2$ (unit radius), the nonzero lattice vectors are $\pm 1, \pm 2, \ldots$, each contributing $|m|^{-2s}$. By symmetry $m \mapsto -m$:

$$\varepsilon_s^c(V_\mathbb{Z}) = 2\sum_{m=1}^{\infty} m^{-2s} = 2\,\zeta(2s).$$

For the even lattice with $Q(m) = 2m^2$ (self-dual radius): $\varepsilon_s^c = 2\cdot(2)^{-s}\,\zeta(2s) = 2^{1-s}\zeta(2s)$. The alternate convention with $\varepsilon_s^1(R{=}1) = 4\zeta(2s)$ uses both positive and negative lattice vectors with unit norm. □

Remark 3.3 (Functional equation)*.* The Epstein zeta function of a rank-r lattice Λ satisfies the functional equation

$$\pi^{-s}\,\Gamma(s)\,E_\Lambda(s) = |\det G|^{-1/2}\,\pi^{-(r/2-s)}\,\Gamma(r/2-s)\,E_{\Lambda^*}(r/2-s),$$

where G is the Gram matrix of Λ and Λ^* is the dual lattice. For the self-dual lattice $\Lambda = \Lambda^*$ at rank 1, this reduces to the functional equation of $\zeta(2s)$. The constrained Epstein zeta inherits this functional equation from the standard Epstein zeta.

3.3. **Higher-rank decomposition.** At rank $r \geq 2$, the Epstein zeta function is no longer a product of Riemann zeta functions. The Hecke decomposition provides the correct generalisation.

Proposition 3.4 (Hecke decomposition)**.** *For a lattice vertex algebra V_Λ of rank r with Gram matrix G, the constrained Epstein zeta function admits a decomposition*

$$\varepsilon_s^c(V_\Lambda) = \sum_j c_j\, L(s, \chi_j), \tag{3.3}$$

where $L(s, \chi_j)$ are L-functions attached to Hecke characters χ_j of the adelic group of the lattice, and $c_j \in \mathbb{Q}$ are algebraic coefficients determined by the lattice geometry.

Remark 3.5 (Lattice theta function)*.* The lattice theta function $\Theta_\Lambda(\tau) = \sum_{\ell \in \Lambda} q^{Q(\ell)/2}$ is a modular form for a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$, and the constrained Epstein zeta is its Mellin transform: $\varepsilon_s^c(V_\Lambda) = (2\pi)^{-s}\,\Gamma(s) \int_0^\infty (\Theta_\Lambda(iy) - 1)\, y^{s-1}\, dy$. The subtraction of 1 removes the zero vector, implementing the bar-complex augmentation at the level of L-functions.

The Niemeier lattice vertex algebras V_Λ at rank 24 have modular characteristic $\kappa = 12$, independent of the root system. The lattice theta function discriminates between the 24 Niemeier lattices; the shadow tower, which depends only on κ, does not. The constrained Epstein zeta function $\varepsilon_s^c(V_\Lambda)$ carries this finer arithmetic.

4. Depth decomposition of the shadow L-function

4.1. **The shadow L-function.** The shadow obstruction tower $\{S_r(\mathcal{A})\}_{r \geq 2}$ is the sequence of degree-r projections of the Maurer–Cartan element $\Theta_\mathcal{A} \in \mathrm{MC}(\mathfrak{g}_\mathcal{A}^{\mathrm{mod}})$, with $S_2 = \kappa(\mathcal{A})$.

Definition 4.1 (Shadow L-function)**.** For a chirally Koszul algebra $\mathcal{A}$, the *shadow L-function* is the Dirichlet series

$$L_\mathcal{A}^{\mathrm{sh}}(s) := \sum_{r \geq 2} S_r(\mathcal{A})\, r^{-s}. \tag{4.1}$$

Example 4.2 (Shadow L-functions by family)**.** The four depth classes produce qualitatively different L-functions.

(a) *Class G* (Heisenberg, $r_{\max} = 2$): $S_r = 0$ for $r \geq 3$, so $L_{\mathcal{H}_k}^{\mathrm{sh}}(s) = k \cdot 2^{-s}$. A single term: the "zeta function" of a point.
(b) *Class L* (affine Kac–Moody, $r_{\max} = 3$): $L_{V_k(\mathfrak{g})}^{\mathrm{sh}}(s) = \kappa \cdot 2^{-s} + S_3 \cdot 3^{-s}$. Two terms: the zeta function of a segment.
(c) *Class C* ($\beta\gamma$, $r_{\max} = 4$): Three terms; the quartic shadow coefficient S_4 controls the discriminant $\Delta = 8\kappa S_4$.
(d) *Class M* (Virasoro, $\mathcal{W}_N$; $r_{\max} = \infty$): The full Dirichlet series is infinite. For Virasoro: $S_4(\mathrm{Vir}_c) = 10/[c(5c + 22)]$.

4.2. **Analytic properties.**

Proposition 4.3 (Poles and zeros of $L^{\mathrm{sh}}_{\mathcal{A}}$)**.** *For a class-M algebra with shadow coefficients $S_r = O(r^{-\alpha})$ for some $\alpha > 0$, the shadow L-function $L^{\mathrm{sh}}_{\mathcal{A}}(s)$ has the following properties:*

(i) (Convergence.) *$L^{\mathrm{sh}}_{\mathcal{A}}(s)$ converges absolutely for $\Re(s) > 1 + 1/\alpha$.*
(ii) (Poles.) *$L^{\mathrm{sh}}_{\mathcal{A}}(s)$ has poles at $s = 1$ and $s = 2$, arising from the leading shadow coefficients $S_2 = \kappa$ and S_3.*
(iii) (Trivial zeros.) *At negative integers $s = -n$ ($n \geq 0$), the shadow L-function has trivial zeros arising from the polynomial vanishing of $\sum r^n S_r$ at finitely many exceptional values of the central charge.*

The shadow L-function is *not* in the Selberg class: it has no Euler product (the coefficients S_r are not multiplicative), and the growth of $|S_r|$ for class-M algebras violates the Ramanujan bound $|a_r| \leq r^{1/2+\varepsilon}$. The arithmetic of the shadow tower is not automorphic; it is a new kind of L-function attached to vertex algebras.

4.3. **Depth stratification.** The four depth classes **G/L/C/M** induce a filtration on the space of shadow L-functions.

Definition 4.4 (Depth filtration)**.** The *depth-d truncation* of the shadow L-function is

$$L^{\mathrm{sh},\leq d}_{\mathcal{A}}(s) := \sum_{r=2}^{d} S_r(\mathcal{A})\, r^{-s}. \tag{4.2}$$

The *depth defect* is $\delta L^{\mathrm{sh}}_{\mathcal{A}}(s) := L^{\mathrm{sh}}_{\mathcal{A}}(s) - L^{\mathrm{sh},\leq d}_{\mathcal{A}}(s) = \sum_{r>d} S_r\, r^{-s}$.

For class G: $\delta L^{\mathrm{sh},\leq 2}_{\mathcal{A}} = 0$ (no defect). For class L: $\delta L^{\mathrm{sh},\leq 3}_{\mathcal{A}} = 0$. For class C: $\delta L^{\mathrm{sh},\leq 4}_{\mathcal{A}} = 0$. For class M: $\delta L^{\mathrm{sh},\leq d}_{\mathcal{A}} \neq 0$ for all finite d; the defect is genuinely infinite.

The discriminant $\Delta(\mathcal{A}) = 8\kappa(\mathcal{A})\, S_4(\mathcal{A})$ vanishes if and only if $\Delta = 0$, separating the finite-tower algebras (classes G, L with $S_4 = 0$) from the infinite-tower algebras (classes C, M with $S_4 \neq 0$ when $\kappa \neq 0$). The depth decomposition of $L^{\mathrm{sh}}_{\mathcal{A}}$ is therefore controlled by a single invariant: Δ.

5. Verlinde polynomials from ordered chiral homology

5.1. **Recovery of the Verlinde formula.** At integer level $k \geq 1$, the quantum group parameter $q = e^{2\pi i/(k+2)}$ is a root of unity, the representation category of $\widehat{\mathfrak{sl}}_2$ at level k truncates to $k+1$ integrable modules, and the ordered chiral homology at each genus computes a finite-dimensional invariant. Write $n = k + 2$ for the shifted level.

Proposition 5.1 (Verlinde formula from ordered chiral homology)**.** *Let $k \geq 1$ be a positive integer, and let $S_{jl} = \sqrt{2/n}\, \sin\bigl(\pi(j+1)(l+1)/n\bigr)$ be the modular S-matrix for $\widehat{\mathfrak{sl}}_2$ at level k, where $j, l \in \{0, 1, \ldots, k\}$.*

(i) (Genus 0.) $\dim H^0\big(\mathcal{M}_{0,0},\, V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^2 = 1$.
(ii) (Genus 1.) $\dim H^0\big(\mathcal{M}_{1,0},\, V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^0 = k+1$.
(iii) (General genus.) $Z_g(k) := \dim H^0\big(\mathcal{M}_{g,0},\, V_k(\mathfrak{sl}_2)\big) = \sum_{j=0}^{k} S_{0j}^{2-2g}$.
(iv) (Handle attachment.) $Z_{g+1} = \sum_{j=0}^{k} S_{0j}^{-2} \cdot S_{0j}^{2-2g} = \sum_{j=0}^{k} S_{0j}^{2-2(g+1)}$.
(v) (Separating factorization.) *Under* $\Sigma_g \rightsquigarrow \Sigma_{g_1} \cup \Sigma_{g_2}$ *with* $g = g_1 + g_2$*:* $Z_g = \sum_{j=0}^{k} S_{0j}^{2-2g_1} \cdot S_{0j}^{2-2g_2} \cdot S_{0j}^{-2}$.

Proof. Part (i): the identity $\sum_j S_{0j}^2 = 1$ is the $(0,0)$-entry of $S \cdot S^T = I$, following from the orthogonality relation $\sum_{j=0}^{k} \sin(\pi(j+1)a/n)\sin(\pi(j+1)b/n) = n\,\delta_{ab}/2$. Part (ii): $S_{0j}^0 = 1$ for all j, so $Z_1 = k+1$. Part (iii): the KZB local system at integrable level decomposes into $k+1$ channels labeled by integrable representations; each channel j contributes S_{0j}^{2-2g} to the partition function. Parts (iv) and (v) follow from the bar complex factorization under sewing. □

The first values are:

k	Z_0	Z_1	Z_2	Z_3	Z_4
1	1	2	4	8	16
2	1	3	10	36	136
3	1	4	20	120	800
4	1	5	35	329	3,611
5	1	6	56	784	13,328

Remark 5.2 (F_g versus Z_g). The shadow coefficients F_g and the Verlinde dimensions Z_g are different invariants of the same algebra. The shadow coefficient $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) varies continuously with the level and is a small rational number (e.g., $F_2 = 7/5760$ at $k = 1$); the Verlinde dimension Z_g is a positive integer defined only at integer level (e.g., $Z_2 = 4$ at $k = 1$). The shadow tower is the generic-level ancestor; the Verlinde formula is the root-of-unity specialisation.

5.2. **The Verlinde polynomial family.** The Verlinde dimensions at each genus are polynomials in $n = k+2$.

Theorem 5.3 (Verlinde polynomial family)**.** *For* $\widehat{\mathfrak{sl}}_2$ *at level* $k \geq 1$*, the Verlinde dimension* $Z_g(k) = \sum_{j=0}^{k} S_{0j}^{2-2g}$ *is a polynomial* $P_g(n)$ *of degree* $3(g-1)$ *in* $n = k+2$ *with the following structure.*

(i) (Universal factor.) *For* $g \geq 2$*,*

$$P_g(n) = n^{g-1}(n^2-1) \cdot R_{g-2}(n^2), \tag{5.1}$$

where R_{g-2} *is a polynomial of degree* $g-2$ *in* n^2 *with rational coefficients and positive leading term.*

(ii) (Explicit formulas through $g = 6$.)

$$P_2(n) = \frac{n(n^2-1)}{6}, \tag{5.2}$$

$$P_3(n) = \frac{n^2(n^2-1)(n^2+11)}{180}, \tag{5.3}$$

$$P_4(n) = \frac{n^3(n^2-1)(2n^4+23n^2+191)}{7560}, \tag{5.4}$$

$$P_5(n) = \frac{n^4(n^2-1)(n^2+11)(3n^4+10n^2+227)}{226800}, \tag{5.5}$$

$$P_6(n) = \frac{n^5(n^2-1)(2n^8+35n^6+321n^4+2125n^2+14797)}{2993760}. \tag{5.6}$$

(iii) (Leading asymptotics.) *The leading coefficient as $n \to \infty$ is*

$$P_g(n) \sim \frac{\zeta(2g-2)}{2^{g-2}\,\pi^{2g-2}}\, n^{3(g-1)} \qquad (n \to \infty). \tag{5.7}$$

(iv) (Rational generating function.) *At fixed n,*

$$G_n(x) := \sum_{g\geq 1} Z_g(k)\, x^{g-1} = \sum_{j=1}^{n-1} \frac{1}{1-a_j\, x}, \qquad a_j = \frac{n}{2\sin^2(\pi j/n)}. \tag{5.8}$$

Proof. The cosecant-power representation $Z_g(k) = (n/2)^{g-1} \sum_{j=1}^{n-1} \csc^{2g-2}(\pi j/n)$ follows from $S_{0j} = \sqrt{2/n}\,\sin(\pi j/n)$. The cosecant power sum $\sigma_m(n) = \sum_{j=1}^{n-1} \csc^{2m}(\pi j/n)$ is a polynomial of degree $2m$ in n with rational coefficients: a classical result of Euler, with leading coefficient $\sigma_m(n) \sim (2/\pi)^{2m}\,\zeta(2m)\,n^{2m}$. Parts (i)–(iii) follow. Part (iv): the geometric series $\sum_{g\geq 1} a_j^{g-1} x^{g-1} = 1/(1-a_j x)$ applied term by term gives the rational generating function. The explicit formulas (5.2)–(5.6) are obtained by polynomial interpolation and verified by direct S-matrix summation. □

Remark 5.4 (Zeta values in the Verlinde tower). The leading coefficient $[n^{3g-3}]P_g = \zeta(2g-2)/(2^{g-2}\pi^{2g-2})$ connects the Verlinde tower to zeta values at even positive integers:

$$\frac{\zeta(2)}{2^0\,\pi^2} = \frac{1}{6}, \quad \frac{\zeta(4)}{2\,\pi^4} = \frac{1}{180}, \quad \frac{\zeta(6)}{4\,\pi^6} = \frac{1}{7560}, \quad \frac{\zeta(8)}{8\,\pi^8} = \frac{1}{226800}.$$

These are the denominators of P_2, P_3, P_4, P_5 respectively. The arithmetic of level-k conformal blocks encodes zeta values: this is a "Verlinde zeta correspondence."

Remark 5.5 (Structural observations). Three features of the polynomial family (5.1):

(a) The factor $n^2 - 1 = (k+1)(k+3)$ vanishes at $k = -1$ and $k = -3$, reflecting the boundary of the integrable range. The factor n^{g-1} vanishes at $n = 0$ ($k = -2$, the critical level), where the Verlinde formula degenerates.

(b) $P_2(n) = \binom{n+1}{3}$ is the unique genus at which Z_g is a single binomial coefficient (the tetrahedral numbers).
(c) The factor $(n^2 + 11)$ appears in both P_3 and P_5, suggesting a recurrence in the polynomials R_{g-2}.

6. The Böcherer bridge: genus-2 Siegel modular forms

6.1. **From the Maurer–Cartan element to Siegel modular forms.** At genus $g \geq 2$, the graph amplitudes produced by the bar complex are modular forms for $\mathrm{Sp}(2g, \mathbb{Z})$. The genus-2 case is the first that connects the Maurer–Cartan element to the arithmetic of Siegel modular forms through a deep conjecture in analytic number theory.

The genus-2 bar complex on Σ_2 with Siegel period matrix $\Omega \in \mathbb{H}_2 = \{Z \in M_2(\mathbb{C}) : Z = Z^t,\ \mathrm{Im}\, Z > 0\}$ involves the prime form $E(z, w)$ on Σ_2 and the matrix $(\mathrm{Im}\,\Omega)^{-1}$ (a 2×2 matrix, not a scalar; at $g = 1$ this reduces to $1/\mathrm{Im}(\tau)$). The bar propagator is $\eta_{ij}^{(2)} = d \log E(z_i, z_j)$, weight 1.

6.2. **The Hecke decomposition at genus 2.** The Hecke operators T_p act on the space $M_{\mathcal{A}}$ of activated amplitudes, decomposing $M_{\mathcal{A}} = \bigoplus_\chi M_\chi$ into generalised eigenspaces. The *arithmetic packet connection*

$$\nabla_{\mathcal{A}}^{\mathrm{arith}} = d - \bigoplus_\chi \frac{\Lambda'_\chi(s)}{\Lambda_\chi(s)} (\mathrm{Id}_{M_\chi} + N_\chi)\, ds \tag{6.1}$$

is a flat meromorphic connection on $M_{\mathcal{A}} \otimes \mathcal{O}_{\mathbb{C}}$ over the spectral s-line, where $\Lambda_\chi(s)$ is the completed L-function of the eigenform and N_χ is the nilpotent part. The singular divisor depends only on the semisimple part $M_{\mathcal{A}}^{\mathrm{ss}}$: the arithmetic is semisimple, the homotopy defect is unipotent.

6.3. **The Saito–Kurokawa lift and the Böcherer conjecture.** For even unimodular lattice vertex algebras, the genus-2 amplitude is a Siegel modular form for $\mathrm{Sp}(4, \mathbb{Z})$, and the Böcherer conjecture provides a bridge to central L-values.

Conjecture 6.1 (Böcherer bridge, after Furusawa–Morimoto)**.** *For the Leech lattice vertex algebra $V_{\Lambda_{24}}$, the genus-2 Siegel modular form produced by the bar complex is the Saito–Kurokawa lift*

$$\chi_{12} = \mathrm{SK}(f_{22}), \tag{6.2}$$

where $f_{22} \in S_{22}(\mathrm{SL}_2(\mathbb{Z}))$ is the unique weight-22 cuspidal eigenform and SK *is the Saito–Kurokawa lifting from elliptic to Siegel modular forms of degree 2.*

The Maurer–Cartan-derived second Fourier coefficient $c_2 \approx -1.918 \times 10^{-6} \neq 0$ is the first nonzero central L-value produced by the shadow obstruction tower. The Böcherer conjecture (now a theorem of Furusawa–Morimoto for $\mathrm{GSp}(4)$ [6]) relates the square of the Fourier coefficient to a

central L-value:

$$|c_2(F)|^2 \sim \frac{L(1/2, F \otimes \chi_D)}{L(1, F, \mathrm{Ad})}, \tag{6.3}$$

where F is a Siegel eigenform and χ_D is a fundamental discriminant character. The MC element $\Theta_{\mathcal{A}}$ connects the bar complex to this arithmetic: the genus-2 amplitude is a Siegel modular form, its Fourier coefficients encode central L-values, and the Böcherer relation computes them from the shadow data.

Remark 6.2 (The Weil analogy). The correspondence between the MC equation and Weil's proof of the Riemann hypothesis for curves over finite fields is structural, not literal:

Weil proof for $C/\mathbb{F}_q$	**MC equation for $\mathcal{A}$**
Curve $C/\mathbb{F}_q$	Chirally Koszul $\mathcal{A}$
Frobenius Frob_q	MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$
$\mathrm{Frob}^2 = q$	$D_{\mathcal{A}}^2 = 0$
Étale cohomology $H^i_{\text{ét}}$	Cyclic deformation $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$
Weight filtration	Degree filtration $\Theta_{\mathcal{A}}^{\leq r}$
Eigenvalues α_i on H^1	Spectral atoms λ_j
$\|\alpha_i\| = q^{1/2}$	Ramanujan bound $\|a_{f_j}(p)\| \leq 2p^{(k-1)/2}$

For even unimodular lattice VOAs, the chain MC equation $\Rightarrow$ prime-local closure $\Rightarrow$ CPS automorphy $\Rightarrow$ Ramanujan bound is the analogue of the Weil mechanism. Deligne's theorem supplies the bound; the MC framework supplies the Hecke decomposition.

7. The constrained Epstein zeta and the Dirichlet–sewing lift

7.1. **The Dirichlet–sewing lift.** The genus-1 connected free energy $F_1^{\mathrm{conn}}(\mathcal{A}; q) = -\log\det(1 - K_q)$ encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. The *Dirichlet–sewing lift* is

$$S_{\mathcal{A}}(u) = \sum_{N \geq 1} a_{\mathcal{A}}(N)\, N^{-u} = \frac{(2\pi)^u}{\Gamma(u)} \int_0^\infty F_{\mathcal{A}}^{\mathrm{conn}}(e^{-2\pi y})\, y^{u-1}\, dy. \tag{7.1}$$

This is a Dirichlet series whose analytic structure is controlled by $\Theta_{\mathcal{A}}$.

Example 7.1 (Sewing Dirichlet series by family)**.**

$$S_{\mathcal{H}}(u) = \zeta(u)\,\zeta(u+1), \tag{7.2}$$

$$S_{V_k(\mathfrak{g})}(u) = \dim(\mathfrak{g}) \cdot \zeta(u)\,\zeta(u+1), \tag{7.3}$$

$$S_{\mathrm{Vir}}(u) = \zeta(u+1)\,(\zeta(u) - 1), \tag{7.4}$$

$$S_{\mathcal{W}_N}(u) = \zeta(u+1)\Big((N-1)\,\zeta(u) - \sum_{j=1}^{N-1} H_j(u)\Big), \tag{7.5}$$

where $H_j(u) = \sum_{m=1}^{j} m^{-u}$ are partial harmonic sums.

Remark 7.2 (The Heisenberg base point). The Heisenberg sewing Dirichlet series $S_{\mathcal{H}}(u) = \zeta(u)\,\zeta(u+1)$ is a product of two shifted Riemann zeta functions, connecting the sewing amplitudes to the convolution structure of divisor sums. Every other family's sewing series decomposes as the Heisenberg core plus a defect: the affine Kac–Moody series is $\dim(\mathfrak{g})$ copies of Heisenberg, the Virasoro series subtracts the constant term 1, and $\mathcal{W}_N$ has $N-1$ Heisenberg copies minus partial harmonic corrections.

7.2. **The three-tier Euler–Koszul classification.** The *Euler defect* $D_{\mathcal{A}}(u) := S_{\mathcal{A}}(u)/(|W(\mathcal{A})|\,\zeta(u)\,\zeta(u+1))$, where $|W(\mathcal{A})|$ is the number of strong generators, classifies chiral algebras into three tiers.

Tier	**Condition**	**Families**
Exact Euler–Koszul	$D_{\mathcal{A}}(u) \equiv 1$	$\mathcal{H}$, $V_k(\mathfrak{g})$, $\beta\gamma$
Finitely defective	$1 - D_{\mathcal{A}}(u)$ finite harmonic ratio	Virasoro, $\mathcal{W}_N$
Non-Euler–Koszul	$S_{\mathcal{A}}$ decomposes into Hecke L-functions	Lattice VOAs with $h(D) \geq 2$

The Euler–Koszul class $\mathrm{ek}(\mathcal{A}) = \max_i(w_i - 1)$, where w_i are the conformal weights of the strong generators, and the shadow depth $r_{\max}(\mathcal{A})$ are independent invariants: character arithmetic and OPE interaction are logically independent axes of complexity.

7.3. **The two-variable L-object.** Rankin–Selberg unfolding against $E^*(\tau, s)$ gives

$$L_{\mathcal{A}}(s,u) = \int_{\mathcal{F}^{\mathrm{reg}}} F_{\mathcal{A}}^{\mathrm{conn}}(\tau)\, E^*(\tau,s)\, y^{u-2}\, dx\, dy, \tag{7.6}$$

where $\mathcal{F}^{\mathrm{reg}}$ is the regularised fundamental domain. The *sewing–Hecke reciprocity theorem* collapses both variables to one:

$$L_{\mathcal{A}}(s,u) = \frac{\Gamma(v)}{(2\pi)^v}\, S_{\mathcal{A}}(v), \qquad v = s + u - 1. \tag{7.7}$$

The scattering poles of E^* at $s = \rho/2$ (nontrivial zeros ρ of ζ) are not poles of $L_{\mathcal{A}}$; the nontrivial zeros appear as structural obstructions to the Euler product, not as poles of the L-object.

7.4. **Constrained Epstein zeta at higher rank.** For a lattice VOA V_Λ of rank r, each shadow depth stratum d contributes $d-1$ critical lines to the meromorphic continuation of $\varepsilon_s^c(V_\Lambda)$, one for each degree stratum beyond the Gaussian base. The nontrivial zeros of the component L-functions are the scattering resonances of the hyperbolic Laplacian on $\mathcal{M}_{1,1}$: frequencies at which the genus-1 sewing amplitude fails to decay.

Remark 7.3 (The finite Miura defect). The principal $\mathcal{W}_N$ character is $\chi_{\mathcal{W}_N}(q) = \prod_{n\geq 2}(1-q^n)^{-\min(n-1,N-1)}$; the Heisenberg character is $\chi_{\mathcal{H}}(q) =$

$\prod_{n\geq 1}(1-q^n)^{-1}$. The ratio

$$\chi_{\mathcal{W}_N}(q) = \chi_{\mathcal{H}}(q)^{N-1} \cdot \prod_{m=1}^{N-1}(1-q^m)^{N-m}$$

is a finite product: the *Miura defect*. All genus-1 arithmetic content in $\mathcal{W}_N$ beyond Heisenberg is carried by this finite product. The prime-side Dirichlet analogue is $D^{\mathrm{prime}}_{\mathcal{W}_N}(u) = -\zeta(u+1)\sum_{m=1}^{N-1}(N-m)\,m^{-u}$; the packet connection splits into $(N-1)$ Heisenberg copies plus a defect governing modes $m = 1, \ldots, N-1$.

Remark 7.4 (Shadow-moduli resolution). The shadow-moduli map $t\colon \mathfrak{H} \to \mathbb{C}$ solves $\prod_j (1-\lambda_j\, t(q))^{c_j} = \det(1-K_q^{\mathrm{conn}})$ with spectral atoms λ_j and multiplicities c_j. The connected free energy factorises as $F_1^{\mathrm{conn}}(q) = G_{\mathcal{A}}(t(q))$ with

$$G_{\mathcal{A}}(t) = \int \log(1-\lambda t)\, d\rho_{\mathcal{A}}(\lambda),$$

determined by the spectral measure $\rho_{\mathcal{A}}$ alone. The moments $\mu_r = \int \lambda^r\, d\rho_{\mathcal{A}}$ satisfy the MC recursion degree by degree: the bar-complex differential equations control the spectral resolution of the genus-1 amplitude.

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